

AUNTOR CHAKMA

[Portfolio](#) | [GitHub](#) | [LinkedIn](#) | auntorchakma@gmail.com | [Google Scholar](#)

SKILLS

- **Machine Learning & Deep Learning:** PyTorch, CNN architectures, Transformer architectures, Attention mechanisms, Federated Learning, Diffusion Models
- **Languages & Frameworks:** Python, PyTorch, Scikit-learn, FastAPI, OOP
- **GenAI & LLM Tools:** LangChain, Groq API, OpenAI API, Hugging Face Transformers
- **AI Domains:** Computer Vision (image captioning, vision-language models)
- **Data & Vector Search:** ChromaDB, FAISS, Pandas, NumPy, Matplotlib, Jupyter Notebook
- **Core CS:** Data Structures and Algorithms, Database Management Systems, Operating Systems, Computer Architecture
- **Infrastructure & Dev:** Git/GitHub, Docker, REST APIs
- **Research:** Experimental methodology, academic writing, manuscript preparation

PROJECTS

AI Customer Support Chatbot (RAG)

[Live](#)

A RAG-based chatbot for an e-commerce store that answers questions about products, shipping, and policies with grounded source references.

- Built an end-to-end RAG pipeline with LangChain and ChromaDB, supporting document ingestion from PDF, DOCX, Markdown, and URL sources with a provider-switching interface across Groq, OpenAI, and Ollama
 - Deployed on Streamlit Community Cloud with keyword-based NLP retrieval and cached resources for low-latency responses
- Stack:** Python, LangChain, ChromaDB, FastAPI, Streamlit, Groq, Ollama

VisualMind AI

[Live](#)

A multimodal computer vision application that captions uploaded images and answers follow-up questions using a vision-language model.

- Chained Hugging Face BLIP-large for image captioning with Groq Llama 4 Scout for visual analysis, maintaining full conversation context across multi-turn follow-up questions
 - Maintained session-level state with per-session token usage and query count tracking; packaged the application for deployment on Hugging Face Spaces
- Stack:** Python, Streamlit, Groq API, Hugging Face Inference API, Pillow, Docker

HireTrack AI

[Live](#)

A job application pipeline with local ATS resume scoring, LLM-generated cover letters, and semantic job discovery via Exa AI.

- Built a local ATS scoring engine using OOP design patterns and ML-based feature weighting across skill match, experience, education, and keyword overlap; exposed job search, scoring, and skill-gap analysis
 - Integrated FAISS vector search for semantic similarity using text embeddings, multi-provider LLM routing, and Exa AI for job discovery; deployed the full stack with Docker Compose
- Stack:** Python, FastAPI, FAISS, Scikit-learn, Exa AI, Render, Docker

EXPERIENCE

Research Assistant

May 2026 – Present

ELITE Research Lab LLC, Queens, NY, USA (Remote)

Junior Research Assistant

Apr 2026 – May 2026

ELITE Research Lab LLC, Queens, NY, USA (Remote)

AI Research Intern

Jan 2026 – Apr 2026

ELITE Research Lab LLC, Queens, NY, USA (Remote)

Focus: Multimodal and Federated Learning

- Researched multimodal and federated learning architectures with a focus on distributed and cross-modal model development
- Designed and executed model training pipelines covering data preprocessing, optimization, and performance evaluation

- Authored academic manuscripts and led peer-reviewed publication preparation as primary author

Stack: Python, PyTorch, Hugging Face Transformers, Pandas, NumPy

Research Collaborator

Aug 2025 – Dec 2025

ELITE Research Lab LLC, Queens, NY, USA (Remote)

- Implemented Stable Diffusion 2.1 with ControlNet-Canny conditioning to generate synthetic fundus images, addressing class imbalance in a seven-stage severity classification task
- Developed a stacking ensemble of five CNN architectures achieving 91.65% accuracy; integrated Grad-CAM and Grad-CAM++ explainability methods to support clinical interpretability of model decisions
- Authored academic manuscripts as primary author and contributed to peer-reviewed publication preparation

Research Assistant

Dec 2024 – Jun 2025

Department of Computer Science and Engineering

Rangamati Science and Technology University, Bangladesh

Supervisor: Md Mynoddin, Assistant Professor, RMSTU

- Collaborated with a research team to simulate and preprocess network traffic datasets, constructing labeled data pipelines for supervised classification of malicious traffic
- Trained and evaluated CNN-based deep learning models for network intrusion and malicious traffic detection under realistic traffic conditions
- Conducted systematic literature review to identify open research gaps in network security classification, contributing to manuscript preparation

Software Engineer Intern

Mar 2024 – Jun 2024

Celoscope, Bangladesh — Backend Development (Java, Spring Boot, MySQL, REST API)

PUBLICATIONS

Peer-Reviewed Conference Paper

- **Co-Author.** Mohajan, S., Hossain, G.M.S., Mynoddin, M., Chakma, R., Mahmud, T., Kanungoe, P., & **Chakma, A.** (2025). A deep learning-based framework for Bangla vehicle license plate detection and recognition. *Communications in Computer and Information Science*, 2683, 220–235. Springer. [Scopus, WoS] [\[Paper\]](#)

Under Review

- **Primary Author.** **Chakma, A.**, Chakma, R., Rahman, D., & Liew, T.H. (2025). An ensemble approach for diabetic retinopathy severity classification with diffusion-based augmentation and explainability. *Emerging Science Journal* [Impact Factor: 5.83; Scopus Q1 (SJR: 0.734)]. Submitted March 2025.
- **Primary Author.** **Chakma, A.**, Moon, F.T.Z., Jahan, F., Morol, M.K., Liew, T.H., Nandi, D., Jubair, A.A., & Rahman, M. (2025). Multimodal federated learning with privacy-preserving mechanisms in healthcare: a mini review of convergence, challenges, and future directions. *Frontiers in Digital Health* [Impact Factor: 3.81; Scopus Q1 (SJR: 1.07); WoS ESCI]. Submitted April 2025.

Thesis

- **Attentive TabNet Framework for Multiclass Intrusion Detection in Internet of Medical Things.** Developed an intrusion detection system for IoMT environments using TabNet with attention mechanisms and GAN-based synthetic data generation to address class imbalance in network security classification.

EDUCATION

Bachelor of Science in Computer Science and Engineering

Sep 2018 – Feb 2025

Rangamati Science and Technology University, Bangladesh

CGPA: 3.50/4.00

Higher Secondary Certificate (HSC)

2015 – 2017

Notre Dame College, Dhaka, Bangladesh

GPA: 5.00/5.00

Secondary School Certificate (SSC)

2015

Laker's Public School and College

GPA: 5.00/5.00

EXTRACURRICULAR ACTIVITIES

- Solved over 100 problems on LeetCode, Codeforces, and URI Online Judge covering data structures and algorithms
- Contributed to a 6-member team that built the official RMSTU International Conference on Bioinformatics website using the MERN stack; handled REST API development and web vulnerability assessment
- Third Place, Intra-University Hackathon (2023)

ACHIEVEMENTS AND AWARDS

- Merit Scholarship, Rangamati Science and Technology University (2023–2024, 2024–2025)
- Chittagong Hill Tracts Development Board Scholarship (2018, 2023, 2024, 2025)
- Rangamati Hill District Council Scholarship (2023, 2025)
- General Merit Scholarship in SSC and HSC by DSHE, Government of Bangladesh (2015, 2017)
- Third Place, Intra-University Hackathon (2023)

REFERENCES

Md Mynoddin

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